THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · EASY LEVEL CHAPTER TEST SET 2

Chapter 9: **Propositions and their Converses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The statement “Every odd number is prime” is: 1

(a) true only for even numbers **(b)** true

(c) true only for prime numbers **(d)** false

2. “All multiples of 10 end in 0” can be written in if–then form as: 1

(a) If not (a number is a multiple of 10), then it ends in 0.

(b) A number is a multiple of 10 or it ends in 0.

(c) If a number is a multiple of 10, then it ends in 0.

(d) If it ends in 0, then a number is a multiple of 10.

3. The number of factors of 12 is: 1

(a) 8 **(b)** 5 **(c)** 6 **(d)** 7

4. The converse of the proposition “If a number is a multiple of 15, then it is a multiple of 5” is: 1

(a) If it is not true that a number is a multiple of 15, then it is not true that it is a multiple of 5.

(b) If it is not true that it is a multiple of 5, then a number is a multiple of 15.

(c) If a number is a multiple of 5, then it is a multiple of 15.

(d) If a number is a multiple of 15, then it is a multiple of 5.

5. Which of the following numbers is divisible by 9? 1

(a) 4537 **(b)** 4533 **(c)** 4530 **(d)** 4536

6. The proposition “If two lines are parallel, then the alternate interior angles made by a transversal are equal” is true. Its converse “If the alternate interior angles made by a transversal are equal, then two lines are parallel” is: 1

(a) It cannot be decided **(b)** True

(c) True only for some values **(d)** False

7. In “If a parallelogram is a rectangle, then its diagonals are equal”, the hypothesis is: 1

(a) its diagonals are equal

(b) a parallelogram is a rectangle

(c) a parallelogram is a rectangle, then its diagonals are equal

(d) it is not true that a parallelogram is a rectangle

8. In “If two lines are parallel, then the alternate interior angles made by a transversal are equal”, the conclusion is: 1

(a) two lines are parallel

(b) it is not true that the alternate interior angles made by a transversal are equal

(c) two lines are parallel, then the alternate interior angles made by a transversal are equal

(d) the alternate interior angles made by a transversal are equal

9. The statement “Every multiple of 3 is odd” is: 1

(a) true only for even numbers **(b)** true

(c) false **(d)** true only for prime numbers

10. “Every equilateral triangle is isosceles” can be written in if–then form as: 1

(a) A triangle is equilateral or it is isosceles.

(b) If not (a triangle is equilateral), then it is isosceles.

(c) If a triangle is equilateral, then it is isosceles.

(d) If it is isosceles, then a triangle is equilateral.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Write “Multiples of 5 end in 0 or 5” in if–then form and name the hypothesis. 2

12. Find the number of factors of 15. Is 15 a perfect square? 2

13. Write the converse of “If a triangle is right-angled, then the square of its longest side equals the sum of the squares of the other two sides”. Is it true? 2

14. Write “A rhombus is a parallelogram” in if–then form and name the hypothesis. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the converse of “If a number ends in 5, then it is divisible by 5”. Is the converse true? Justify. 3

16. Show that the statement “$2n + 1$ is prime for every natural number n” is false by giving a counterexample. 3

17. Is a triangle with sides 8 cm, 15 cm and 17 cm right-angled? Use the converse of the Baudhāyana–Pythagoras theorem. 3

OR

Write the converse of “If a number is prime and greater than 2, then it is odd”. Is the converse true? Justify. 3

18. Show that the statement “$n^2 - n + 11$ is prime for all natural numbers n” is false by giving a counterexample. 3

OR

Is a triangle with sides 7 cm, 24 cm and 26 cm right-angled? Use the converse of the Baudhāyana–Pythagoras theorem. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Verify with 9 and 13 that the sum of two odd numbers is even, and then prove it for any two odd numbers. 5

OR

Prove that the product of two consecutive integers is even. Verify it for 13 and 14. 5

20. Verify with 15 and 21 that the sum of two odd numbers is even, and then prove it for any two odd numbers. 5

OR

Prove that the product of two consecutive integers is even. Verify it for 14 and 15. 5

END OF QUESTION PAPER — **Total: 40 Marks**